Solution Requirements

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The system shall allow users to log in securely using valid credentials.	High
2	Authorization levels	The system shall provide role-based access, where users can submit loan details and administrators can manage data and monitor the system.	High
3	External interfaces	The system shall connect to the machine learning model and database to process applicant data and generate predictions.	Medium

4	Transactions processing	The system shall collect applicant information, process the data through the prediction model, and generate loan eligibility results in real time.	High
5	Reporting	The system shall generate prediction results and allow administrators to view applicant records and system reports.	Medium
6	Business rules, etc.	The system shall determine loan eligibility based on factors such as income, loan amount, credit history, employment status, and other applicant details	High
7	Compliance to laws or regulations	The system shall protect user data and comply with applicable data privacy and security standards by ensuring confidentiality and secure data handling.	Medium
8	*Other*	The system shall provide a simple and user-friendly interface for entering applicant details and viewing prediction results.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**

1	Performance & Speed	The system shall process applicant data and display loan eligibility predictions quickly and efficiently.	*Response time < 5 seconds*
2	Scalability	The system shall handle an increasing number of users and prediction requests without performance degradation.	*Supports up to 500 concurrent users.*
3	Security & Data Privacy	The system shall protect user information through secure authentication and data encryption mechanisms.	*Secure login and encrypted data*
4	Reliability & Availability	The system shall be available to users with minimal downtime and provide consistent prediction services.	*99% system uptime per month.*
5	Usability & Accessibility	The system shall provide an intuitive, easy-to-use interface that can be accessed from different devices and browsers.	*Prediction completed in < 5 minutes*
6	*Other*	The system shall be easy to maintain, update, and deploy on different platforms.	Modular and documented code with compatibility across Windows and Linux environments.